

Documentation for "tri front" Surface Tag

1. Purpose of the "tri front" Surface Tag

The "tri front" surface tag is used to indicate **triangular surfaces** in the wing structure where **one of the nodes belongs to the front spar**. This updated naming convention provides clarity regarding the geometric relationship of the triangular surface to the front spar, improving traceability and accuracy in the documentation.

2. Description of "tri front" Surfaces

- **Shape:** Triangular.
- **Key Feature:** One of its nodes is specifically located on the **front spar**.
- **Use Case:** These surfaces typically close the structure near the front spar and are used to connect the front spar to other structural elements (e.g., stringers or ribs).
- **Examples:**
 - Final triangular surfaces connecting the front spar and stringer nodes.
 - Surfaces generated during irregular stringer modeling in zones where the geometry transitions into a triangular shape.

3. Key Metadata and Properties

Each "tri front" surface is defined by:

- **node_1 (local_id):** The first corner of the triangle.
- **node_2 (local_id):** The second corner of the triangle.
- **node_3 (local_id):** The third corner of the triangle, specifically located on the front spar.
- **stringer_1 and stringer_2:** Indices of the stringers involved in the triangular surface.
- **rib_1 and rib_2:** Indices of the ribs involved.
- **tags:** The surface tag "tri front".
- **area:** The computed area of the triangular surface.
- **aspect_ratio:** A measure of the surface's shape quality, used to validate the surface's suitability for structural analysis.

4. Key Changes from "tri" to "tri front"

- **Old Tag Name:** "tri"
 - It did not explicitly indicate the involvement of the front spar, which could cause confusion when multiple triangular surfaces were created in different parts of the structure.
- **New Tag Name:** "tri front"
 - Provides clarity by explicitly identifying triangular surfaces connected to the front spar.

5. Advantages of the Update

- **Improved Clarity:** The tag name now explicitly indicates the geometric location of the surface.
- **Better Documentation:** Helps differentiate between different types of triangular surfaces (e.g., "tri front" vs other potential triangular surfaces in the future).
- **Enhanced Thesis Consistency:** Aligns naming conventions with the structural geometry being modeled.

6. Example Workflow

- **Input:** A triangular surface is created by connecting two stringer nodes (node_1 and node_2) to a front spar node (node_3).
- **Output Metadata Example:**

local_id	node_1	node_2	node_3	stringer_1	stringer_2	rib_1	rib_2	tags	area	aspect_ratio
1	100	101	50	5	6	10	11	"tri front"	1.23	0.8

Thesis Reference Notes

When documenting the "tri front" surfaces in your thesis:

1. **Explain the Purpose:**
 - Clearly describe that "tri front" surfaces are triangular panels connecting the front spar to adjacent stringers and ribs.
2. **Use Visuals:**

- Include diagrams showing a typical "tri front" surface with the front spar node labeled.

3. Provide Metadata:

- Include a table similar to the example above to describe the surface properties (nodes, stringer indices, rib indices, area, and aspect ratio).

4. Clarify Modeling Use Cases:

- Highlight when and why "tri front" surfaces are created (e.g., in irregular zones or near the wing root).